

Meriyam EL HEFIANE

Embedded Software Engineer

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MeriyamELHEFIANE | meriyam-el-hefiane-56400b237

Embedded Software Engineer passionate about **embedded systems, artificial intelligence, IoT, and web and mobile application development**. Strong analytical and problem-solving skills, eager to contribute to innovative embedded, automotive, and software engineering projects.

EXPERIENCE

Embedded Systems & Computer Engineer - Final-Year Internship | OCP – Safi — Feb 2026 – June 2026

- Designed and implemented an **Edge AI** system for real-time prediction and anomaly diagnosis of industrial **SO_2 emissions**, integrating **Machine Learning** models (**Linear Regression**, **Decision Trees**, **Random Forest**, and **XGBoost**) to identify the most accurate predictive model.
- Deployed the solution on a **Raspberry Pi 4** with a **React.js** dashboard connected to **SQL Server**, enabling real-time emissions monitoring, anomaly visualization, weather station supervision, alert management, and automated corrective action recommendations.

Tutor - Internship | Higher School of Technology (EST) – Safi — Jul 2023 – Aug 2023

- Led **Laravel** and **Flutter** workshops for beginner developers, covering fundamental concepts and hands-on practice.

Software Engineer | OCS Tennis Club – Safi — Apr 2023 – May 2023

- Built a responsive **Bootstrap** interface and developed core **Laravel** features (auth, sessions, tournaments, payments, registrations, profile management).

SKILLS

Programming: C, C++, Python, Java, PHP, JavaScript, MATLAB
Embedded Systems & IoT: Arduino, ESP32, ESP8266, Raspberry Pi, VHDL, I2C
Web & Mobile Development: HTML, CSS, Bootstrap, Laravel, React.js, Django REST Framework, Flutter
Databases: MySQL, SQL Server, PostgreSQL, SQLite, Oracle Database
Machine Learning: Supervised Learning, Unsupervised Learning, Reinforcement Learning
Tools: Git, Linux, Postman, Arduino IDE, Proteus, Quartus, MATLAB/Simulink, Talend Open Studio, Power BI, Hadoop

EDUCATION

Master's Degree of Excellence in Computer Engineering and Embedded Systems — 2024 - 2026
Faculty of Sciences, Ibn Zohr University — Agadir, Morocco

Bachelor's Degree of Excellence in Computer Engineering and Embedded Systems — 2023 - 2024
Faculty of Sciences, Ibn Zohr University — Agadir, Morocco

Computer Engineering, University Technology Diploma (DUT) — 2021 - 2023
Higher School of Technology (EST), Cadi Ayyad University — Safi, Morocco

PROJECTS

Autonomous Surveillance Robot

- Developed an autonomous robot using **Arduino Uno** and **ESP-CAM**, with motion and ultrasonic sensors for patrolling, obstacle avoidance, and real-time video surveillance.

SCAM | Connected Cardio-Motor Monitoring Bracelet (*github.com/kawtarknd/SCAM_APP_vf*)

- Engineered a connected **bracelet** (ESP32) with MAX30102/MPU6050 sensors for real-time cardiac and motion monitoring.
- Built a companion mobile app using **Flutter**, **Firebase**, and **Supabase**, integrating **AI-based detection** of cardiac anomalies and stress:
  - Sensor Data → AI Analysis → Anomaly Detection → Secure Transmission.

LANGUAGES

Arabic : Native | **French :** B2 | **English :** B2

CERTIFICATIONS

CCNA: Switching, Routing, and Wireless Essentials, **Internet Society:** Encryption Certificate, **MathWorks:** Image Processing

ACHIEVEMENTS & LEADERSHIP

President - Robotics & Embedded Systems Club | **3rd Prize** - Speed, Code & Innovation - EST 2024
Organizer - TechNovate Hackathon - IT Center of Excellence FSA 2025